

The Design Strength of the Carrier

Why the 24-cell stops at $n=3$ and the 600-cell stops at $n=6$, what that does and does not establish about hydrogen, and how to attack the radial bridge without walking into a circle.

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SECTION ONE

The cutoffs are not mysterious, and that is good news

The hydrogen monograph reports a computed result: the 24-cell exactly represents hydrogen's mode algebra through n equals three and fails at four, and the 600-cell exactly represents it through n equals six and fails at seven. The paper treats those cutoffs as findings and does not explain them. They have an explanation, it is exact, and having it makes the result stronger rather than weaker.

A finite set of points on a sphere is a spherical t -design when averaging any polynomial of degree at most t over the points gives the same answer as integrating it over the whole sphere. It is a cubature rule that is exact up to a stated degree and blind above it. The two polytopes in question have known design strengths, and the strengths were re-derived here from the coordinates rather than cited, by checking every monomial moment against the exact spherical value.

24-cell : 24 points spherical 5-design
600-cell : 120 points spherical 11-design

Now the arithmetic that produces the cutoffs. For the discrete Gram matrix of degree- N harmonics to come out as the identity, the rule must integrate products of two degree- N harmonics, which are polynomials of degree $2N$. So exactness requires $2N$ at most t , that is N at most the floor of t over two. Hydrogen's principal number is N plus one.

24-cell $t = 5 \rightarrow N \leq 2 \rightarrow n \leq 3$ fails at $n = 4$
600-cell $t = 11 \rightarrow N \leq 5 \rightarrow n \leq 6$ fails at $n = 7$

Both cutoffs, exactly, with nothing adjusted. The monograph's computation is correct and its failure points are the design strengths of the polytopes and nothing else. The condition number of one throughout the resolved range, which the monograph also reports, is the same fact seen from another side: a design integrates every direction equally, so no direction in the represented mode space is privileged.

PROVED, and verified from coordinates in scripts/design_strength.py. The monograph should cite this rather than presenting the cutoffs as empirical. A result that was predictable in advance is worth more than one that merely happened.

SECTION TWO

What this establishes, stated narrowly

It is worth being exact about what has been shown, because the composition is elegant and the temptation to overread it is real.

Hydrogen's bound states live on a three-sphere. That is Fock's 1935 result: at fixed negative energy, momentum space compactifies onto S^3 , the hidden $SO(4)$ symmetry becomes manifest, and the bound states become hyperspherical harmonics of degree n minus one. The 600-cell is an excellent design on S^3 . Therefore the 600-cell samples hydrogen's modes exactly up to its design strength.

Two known facts, composed. The composition is correct and it is neat, and it is not a derivation of hydrogen from the framework. What the framework contributes is the claim that E_8 folds to the H_4 carrier in the first place; what the design theory contributes is everything about why the sampling works and where it stops.

the carrier explains the sampling; it does not yet explain the atom

The monograph says this, more or less, in its status table. This paper says it harder, because a referee will otherwise say it for you and less kindly.

SECTION THREE

The radial bridge is two problems, not one

The monograph names the decisive missing theorem: derive the minus one over r operator, or an equivalent, and recover the energies going as minus one over n squared, without fitting. That is the right target. It is also two targets wearing one coat, and separating them shows that one of them is nearly free and the other is the framework's standing problem in a new costume.

The form That the energies go as minus one over n squared. This follows from the S^3 harmonic structure by Fock's construction, with no further input. Requiring the states to be harmonics of integer degree on S^3 quantises the compactification scale, and the energy is minus that scale squared over twice the reduced mass. The inverse square in n is the S^3 structure speaking. If the framework delivers S^3 , it delivers the form.

The scale The Rydberg itself, which needs the coupling. That is the fine structure constant, which is a coupling constant, which is the same thing the framework owes everywhere else and does not owe more here than elsewhere.

So the radial bridge is not one wall. It is one wall the framework may already be standing on the far side of, and one door it has not opened anywhere in the corpus.

SECTION FOUR

The circle, and where to break it

There is a trap in the previous section and it must be named before anyone walks into it. Fock's route runs from the Coulomb potential to the hidden $SO(4)$ symmetry to the three-sphere. The S^3 structure of hydrogen is not independent evidence for anything; it is what the one over r potential looks like after a

change of variables. Deriving one over r from S_3 while having obtained S_3 from one over r is a circle, and a referee will find it in a minute.

So the framework cannot use hydrogen's S_3 and then claim to have explained hydrogen. It has to produce S_3 from its own structure, without touching the Coulomb potential, and then let Fock's construction run backwards.

The attack this paper recommends

The framework has three spatial modes and one closure mode, and it has them for reasons that have nothing to do with atoms. Definition eighteen makes the closure mode the bookkeeping that keeps traversals comparable, and excludes it from the traversable directions. Result sixteen proves it is single.

Three traversable directions with a single closure on them is not an open three-space. It is a three-space in which every traversal returns, and a simply connected three-manifold in which every traversal returns is the three-sphere. That statement is not made anywhere in the corpus, and if it can be made properly then S_3 arrives from the geometry block rather than from the atom, and the circle opens.

three spatial modes, one closure: the closed three-space is S_3

If that holds, the order of the argument reverses and becomes worth something. The carrier gives three spatial modes and a closure. The closure makes the momentum space a three-sphere. Fock's construction, read backwards, converts a three-sphere into a one over r potential and energies going as minus one over n squared. Hydrogen's spectrum would then be a consequence of the geometry block rather than an input to it, and the only remaining quantity would be the coupling.

OPEN, and stated as the single highest-value calculation available to the framework. It is small, it is well posed, it does not require a Lagrangian, and it is falsifiable: either the closure condition on three traversable modes forces the three-sphere or it does not. The corpus should not claim hydrogen until this is done, and should claim it loudly if it is.

SECTION FIVE

Three corrections the hydrogen monograph should carry

Cite the design strengths Present the n at most three and n at most six cutoffs as consequences of the 5-design and 11-design properties, with the floor of t over two arithmetic shown. Predicted beats observed.

Narrow the strongest claim The exact-sampling result is a composition of Fock's S_3 structure with a known cubature property. Say so in the sentence that states it, not only in the status table three sections later.

Split the radial bridge Separate the form, which S_3 may already deliver, from the scale, which is the corpus-wide coupling problem. The monograph currently presents them as one obstacle and thereby makes its position look worse than it is.

None of these weakens the monograph. Two of them strengthen it by making a computed result predictable, and the third shows that the gap it honestly declares is smaller than it looks.